

Case Study 2: The Wellmeadows Hospital Case Study

This case study describes a small hospital called Wellmeadows, which is located in Edinburgh. The *Wellmeadows Hospital* specializes in the provision of health care for elderly people. Listed below is a description of the data recorded, maintained and accessed by the hospital staff to support the management and day-to-day operations of the *Wellmeadows Hospital*.

Wards

The *Wellmeadows Hospital* has 17 wards with a total of 240 beds available for short- and long-stay patients, and an out-patient clinic. Each ward is uniquely identified by a number (for example, ward 11) and also a ward name (for example Orthopedic), location (for example, E Block), total number of beds and telephone extension number (for example, Extn. 7711).

Staff

The *Wellmeadows Hospital* has a Medical Director, who has overall responsibility for the management of the hospital. The Medical Director maintains control over the use of the hospital. The Medical Director maintains control over the use of the hospital resources (including staff, beds and supplies) in the provision of cost-effective treatment for all patients.

The *Wellmeadows Hospital* has a Personnel Officer who is responsible for ensuring that the appropriate number and type of staff are allocated to each ward and the out-patient clinic.

The information stored on each member of staff includes a staff number, name (first) full address, telephone number, date of birth, sex, national insurance number (NIN), position held, current salary, salary scale, qualifications (which includes date of qualification, type, name of institution) and work experience details (which includes the start and finish dates, position, name of organization).

The type of employment contract for each member of staff is also recorded including the number of hours worked per week, whether the member of staff is on a permanent or temporary contract and the type of salary payment (weekly/monthly).

An example of a *Wellmeadows Hospital* form used to record the details of a member of staff called Moira Samuel working in ward 11 is shown in Figure 1.1

Figure 1.1 (*Wellmeadows Hospital member of staff form*)

Wellmeadows Hospital	
Staff Form	
Staff Number: S011	
Personal Details	
First Name: <i>Moira</i>	Last Name: <i>Samuel</i>
Address: <i>49 School Road</i> <i>Broxburn</i>	Sex: <i>Female</i> Date of Birth: <i>30-May-61</i>
Tel. No. <i>01506-45633</i>	NIN <i>WB123423D</i>
Position; <i>Charge Nurse</i>	Allocated: <i>11</i>
Current Salary: <i>18.760</i>	to ward
Salary Scale: <i>1C scale</i>	Hours/Week <i>37.5</i>
Paid weekly or Monthly	Permanent or Temporary (Enter P or T) <i>P</i>

(Enter W or M) M	
Qualification(s)	Work Experience
Type: <i>BSc Nursing Studies</i> Date: <i>12-Jul-87</i> Institution: <i>Edinburgh Unit</i>	Position: <i>Staff Nurse</i> Start Date: <i>23-Jan-90</i> Finish Date: <i>1-May-93</i> Organization: <i>Western Hospital</i>
Note: Please enter additional qualifications/work experience overleaf	

Each ward and the out-patient clinic has a member of staff with the position of Charge Nurse. The Charge Nurse is responsible for overseeing the day-to-day operation of the ward/clinic. The Charge Nurse is allocated a budget to run the ward and must ensure that all resources (staff, beds and supplies) are used effectively in the care of patients. The Medical Director works with the Charge Nurses to ensure the efficient running of the hospital.

A charge Nurse is responsible for setting up a weekly staff rota, and must ensure that the ward/clinic has the correct number and type of staff on duty at any time during the day or night. In a given week, each member of staff is assigned to work an early, late or night shift.

As well as the Charge Nurse, each ward is allocated senior and junior nurses, doctors and auxiliaries. Specialist staff (for example, consultants, physiotherapists) are allocated to several wards or the clinic.

An example of a *Wellmeadows Hospital* report listing the details of the staff allocated to ward 11 is shown in Figure 1.2.

Figure 1.2 (*Wellmeadows Hospital* report listing ward staff)

Page 1		WellmeadowsHospital		Week 9-Jan-96	
		Ward Staff Allocation		beginning	
Ward Number Ward 11		Charge Nurse Moira Samuel			
Ward Name Orthopaedic		Staff Number S011			
Location Block E		Tel Extn 7711			
Staff No.	Name	Address	Tel No	Position	Shift
S098	Carol Cummings	15 High Street Edinburgh	0131-334-5677	Staff Nurse	Late
S123	Morgan Russell	23A George Street Broxburn	01506-67676	Nurse	Late
S167	Robin Plevin	7 Glen Terrace Edinburgh	0131-339-6123	Staff Nurse	Early
S234	Amy O' Donnell	234 prices Street Edinburgh	0131-334-9099	Nurse	Night
S344	Laurence Burns	1 Apple Drive Edinburgh	0131-344-9100	Consultant	Early

Patients

When a patient is first referred to the hospital they are allocated a unique patient number. At this time, additional details of the patient are also recorded including the name (first and last name), address, telephone number, date of birth, sex, marital status, date registered with the hospital and the details of the patient's next-of-kin.

Patient's next-of-kin

The details of a patient's next-of-kin are also recorded and include the next-of-kins's full name, relationship to the patient, address and telephone number.

Local doctors

Patients are normally referred to the hospital for treatment by their local doctor. The details of local doctors are held including their fullname, clinic number, address and telephone number. The clinic number is unique throughout the United Kingdom.

An example of a *Wellmeadows Hospital* patient registration form used to record the details of a patient called Anne Phelps is shown in Figure 1.3.

Figure 1.3 (*Wellmeadows Hospital Patient Registration Form*)

Wellmeadows Hospital Patient Registration Form	
Patient Number: <i>P10234</i>	
Personal Details	
First Name: <i>Anne</i>	Last Name: <i>Phelps</i>
Address: <i>44 North Briges</i>	Sex: <i>Female</i>
<i>Cannonmills</i>	Tel No.: <i>0131-332-4111</i>
<i>Edinburgh, EH1 5GH</i>	Marital Status: <i>Single</i>
DOB: <i>12-Dec-33</i>	Date Registered: <i>21-Feb-95</i>
Next-of-kin Details	
Fullname: <i>James Phelps</i>	Relationship: <i>Father</i>
Address: <i>145 Rowlands Street</i>	
<i>Paisley, PA2 5FE</i>	
Tel No. <i>0141-848-2211</i>	
Local Doctor Details	
Fullname: <i>Dr. Helen Pearson</i>	
Address: <i>22 Cannongate Way,</i>	
<i>Edinburgh, EH1 6TY</i>	
Tel No. <i>0131-332-0012</i>	

Patient appointment

When a patient is referred by his or her doctor to attend the *Wellmeadows Hospital*, the patient is given an appointment for an examination by a hospital consultant.

Each appointment is given a unique appointment number. The details of each patient's appointment are recorded, and include the name and staff number of the consultant undertaking the examination, the date and time of the appointment and the examination room (for example, Room E252).

As a result of the examination, the patient is either recommended to attend the out-patient clinic or is placed on a waiting list until a bed can be found in an appropriate ward.

Out-patients

The details of out-patients are stored and include the patient number, name (first and last name), address, telephone number, date of birth, sex and the date and time of the appointment at the out-patient clinic.

In-patients

The Charge Nurse and other senior medical staff are responsible for the allocation of beds to patients on the waiting list.

The details of patients currently placed in a ward and those on the waiting list for a place on a ward are recorded. This includes the patient number, name (first and last name), address, telephone number, date of birth, sex, marital status, the details of the patient's next-in-kin, the date placed on the waiting list, the ward required, expected duration of stay (in days), date placed in the ward, date expected to leave the ward and the actual date the patient left the ward, when known.

When a patient enters the ward they are allocated a bed with a unique bed number. An example of a *Wellmeadows Hospital* report listing the details of patients allocated to ward 11 is shown in Figure 1.4

Figure 1.4 (*Wellmeadows Hospital* report listing ward patients)

Page 1		WellmeadowsHospital		Week 16-Jan-96			
		Patient Allocation		beginning			
Ward Number		Ward 11		Charge Nurse Moira Samuel			
Ward Name		Orthopaedic		Staff Number S011			
Location		Block E		Tel Extn 7711			
Patient Number	Name	On Waiting List	Expected Stay (Days)	Date Placed	Date Leave	Actual Leave	Bed Number
P10451	Robber Drumtree	12-Jan-96	5	12-Jan-96	17-Jan-96		84
P10480	Steven Parks	12-Jan-96	4	14-Jan-96	18-Jan-96		79
P10563	David Black	13-Jan-96	14	13-Jan-96	27-Jan-96		80
P10604	Ian Thompson	14-Jan-96	10	15-Jan-96	25-Jan-96		87

Patient medication

When a patient is prescribed medication, the details are recorded. This includes the patient's name and number, drug number and name, units per day, method of administration (for example, oral, intravenous (IV)), start and finish date. The medication (pharmaceutical supplies) given to each patient is monitored.

An example of a *Wellmeadows Hospital* report used to record the details medication given to a patient called Robert MacDonald is shown in Figure 1.5

Figure 1.5 (*Wellmeadows Hospital patient's medication report*)

Wellmeadows Hospital Patient Medication Form							
Patient Number: <i>P10034</i> Full Name <i>Ronald MacDonald</i> Bed Number <i>84</i> Ward Number: <i>Ward 11</i> Ward Name: <i>Orthopaedic</i>							
Drug Number	Name	Description	Dosage	Method of Admin	Units per Day	Start Date	Finish Date
10223	Morphine	Pain killer	10mg/ml	oral	50	24-Mar-96	24-Apr-96
10334	Tetracycline	Antibiotic	0.5mg/ml	IV	10	24-Mar-96	17-Apr-96
10223	Morphine	Pain killer	10mg/ml	oral	10	25-Apr-96	2-May-96

Surgical and non-surgical supplies

The *Wellmeadows Hospital* maintains a central stock of surgical (for example, syringes, sterile dressings) and non-surgical (for example, plastic bags, aprons) supplies. The details of surgical and non-surgical supplies include the item number and name, item description, quantity in stock, reorder level and cost per unit. The item number uniquely identifies each type of surgical or non-surgical supply. The supplies used by each ward are monitored.

Pharmaceutical supplies

The hospital also maintains a stock of pharmaceutical supplies (for example antibiotics, pain killers). The details of pharmaceutical supplies include drug number and name, description, dosage, method of administration, quantity in stock, reorder level, cost per unit. The drug number uniquely identifies each type of pharmaceutical supply. The pharmaceutical supplies used by each ward are monitored.

Ward requisitions

When required, the Charge Nurse may obtain surgical, non-surgical and pharmaceutical supplies from the central stock of supplies held by the hospital. This is achieved by ordering supplies for the ward using a requisition form. The information detailed on a requisition form includes a unique requisition number, the name of the member of staff placing the requisition and the number and name of the ward. Also included is the item or drug number, name, description, dosage and method of administration (for drugs only), cost per unit, quantity required and date ordered. When the requisitioned supplies are delivered to the ward, the form must be signed and dated by the Charge Nurse who initiated the order.

An example of a *Wellmeadows Hospital* requisition form used to order supplies or morphine for ward 11 is shown in Figure 1.6.

Suppliers

The details of the suppliers of the surgical, non-surgical and pharmaceutical items are stored. This information includes the supplier's name and number, address, telephone and fax number. The supplier number is unique for each supplier.

Figure 1.6 (*Wellmeadows Hospital ward requisition form*)

Wellmeadows Hospital Central Store Requisition Form						
Requisition Number: 034567712 Ward Number: <i>Ward 11</i> Ward Name: <i>Orthopaedic</i> Requisitioned By: <i>Moira Samuel</i> Requisitioned Date: <i>15-Feb-96</i>						
Item/Drug Number	Name	Description	Dosage (Drugs Only)	Method of Admin	Cost per Unit	Quantity
10223	Morphine	Pain killer	10 mg/ml	Oral	27.75	50

Operations

The following operations are undertaken to ensure that the appropriate information is available to enable the staff to manage and oversee the day-to-day running of the *Wellmeadows Hospital*. Each operation is associated with a specific function within the hospital. These functions are the responsibility of members of staff with particular job titles (positions). The main use or group of users of each operation is given in brackets at the end of the description of each operation.

- (a) Create and maintain records recording the details of members of staff (Personnel Officer)
 - (b) Search for staff who have particular qualifications or previous work experience (Personnel Officer).
 - (c) Produce a report listing the details of staff allocated to each ward (Personnel Officer and Charge Nurse).
 - (d) Create and maintain records recording the details of patients referred to the hospital (all staff)
 - (e) Create and maintain records recording the details of patients referred to the hospital (all staff)
 - (f) Produce a report listing the details of patients referred to the out-patients clinic (Charge Nurse and Medical Director).
 - (g) Create and maintain records recording the details of patients referred to a particular ward (Charge Nurse)
 - (h) Produce a report listing the details of patients currently located in the particular ward (Charge Nurse and Medical Director).
 - (i) Produce a report listing the details of patients currently located in a particular ward (Charge Nurse and Medical Director).
 - (j) Create and maintain records recording the details of medication given to a particular patient (Charge Nurse).
 - (k) Produce a report listing the details of medication for a particular patient (Charge Nurse)
 - (l) Create and maintain record recording the details of suppliers for the hospital (Medical Director)
 - (m) Create and maintain records detailing requisitions for supplies for particular wards (Charge Nurse).
 - (n) Produce a report listing the details of supplies provided to specific wards (Charge Nurse and Medical Director).
- **Provide data model (ER/DfD/UML) and database design (observe normalization rules).**

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